
Autoimmune Disorder Data Analysis and Visualization

Exploring Disease Patterns, Symptoms, Biomarkers, and Patient Demographics through
Data Visualization

Course	Data Science and Visualization (24ADI024)
Project Mode	DATA-VIZ HACK-A-BIT 2026–2027
Week	Week 1 — Team Formation & Data Hunting
Team	Team 12

Team Members

Nivetha S — 25BAD074 | Pooja T — 25BAD077 | Puvi Priya N — 25BAD084 | Vaishnavi D — 25BAD126

GitHub Repository

https://github.com/nivetha486/24ADI204_DSV_Team12

Week 1 Status: Completed

1. Introduction

The first week of the DATA-VIZ HACK-A-BIT project establishes the foundation for the complete Data Science and Visualization workflow. According to the project rule book, Week 1 focuses on **team formation, data hunting and dataset selection, GitHub setup, and an introduction to Pandas and NumPy**. The team selected a healthcare-focused autoimmune disorder dataset and defined the initial analytical direction for exploring disease patterns, symptoms, biomarkers, and patient demographics through visualization.

The project is designed as an end-to-end visual storytelling exercise. The selected topic is aligned with the rule book requirement to work with a real-world dataset and ultimately transform raw data into meaningful visual insights and an interactive dashboard.

2. Week 1 Objectives

- Form a four-member project team and establish a common project direction.
- Identify and select a suitable real-world healthcare dataset for analysis.
- Define the project theme: autoimmune disorder data analysis and visualization.
- Set up the team's GitHub repository for version control and project documentation.
- Become familiar with Pandas and NumPy as the primary Python tools for data handling and numerical operations.
- Define the initial scope of analysis: disease prevalence, symptoms, biomarkers, laboratory findings, and patient demographics.

3. Team Formation

The project team consists of four members, satisfying the rule book requirement that each team contain 3–4 students. The team will collaborate across data preparation, analysis, visualization, documentation, and dashboard development throughout the project.

S.No.	Team Member	Register Number
1	Nivetha S	25BAD074
2	Pooja T	25BAD077
3	Puvi Priya N	25BAD084
4	Vaishnavi D	25BAD126

4. Data Hunting and Dataset Selection

The team selected a **Comprehensive Autoimmune Disorder Dataset** from the healthcare/clinical data domain. The project presentation identifies the source as Kaggle and describes the dataset as containing **12,500 records and 79 features**, with **Autoimmune Disorder Diagnosis** as the target variable.

4.1 Dataset Characteristics

Attribute	Details
Dataset name	Comprehensive Autoimmune Disorder Dataset
Domain	Healthcare / Clinical Data Science
Source	Kaggle (as stated in the project PPT)
Records	12,500
Features	79
Target	Autoimmune Disorder Diagnosis

4.2 Important Feature Groups

- **Demographics:** Patient ID, Age, Gender.
- **Diagnosis:** Disease diagnosis / autoimmune disorder diagnosis.
- **Symptoms:** Fatigue, joint pain, fever and other recorded clinical symptoms.
- **Laboratory parameters:** Hemoglobin, WBC, Platelets, ESR and CRP.
- **Autoantibodies / biomarkers:** ANA, Anti-dsDNA, Anti-Sm, Anti-TPO and ACPA.
- **Clinical history:** Disease duration and related patient-level indicators.

4.3 Reason for Selecting the Dataset

- It represents a real-world healthcare analytics problem with multiple demographic and clinical dimensions.
- The dataset supports both statistical analysis and visual storytelling.
- Symptoms, biomarkers, laboratory values and demographics provide multiple opportunities for univariate, bivariate and multivariate analysis.
- The topic is suitable for identifying patterns and relationships that can later be communicated through an interactive dashboard.

5. Project Problem and Initial Scope

Autoimmune disorders are difficult to analyze because patients can present with overlapping symptoms and different laboratory and biomarker profiles. The project therefore aims to organize and visualize patient-level information so that disease prevalence, symptom patterns, demographic differences, and biomarker relationships can be explored systematically.

Initial analytical scope

- Study the distribution and prevalence of autoimmune disorder diagnoses.
- Explore age and gender distributions across disease groups.
- Examine symptom occurrence and compare symptoms across diseases.
- Compare laboratory parameters such as Hemoglobin, WBC, Platelets, ESR and CRP.

- Study the relationship between autoimmune biomarkers/autoantibodies and disease diagnosis.
- Use visualizations to identify patterns that can support the final storytelling dashboard.

6. GitHub Setup and Version Control

As required by the Hack-a-BIT rule book, the team established a GitHub repository to maintain project files, documentation and analysis work. The repository is publicly accessible and is named **24ADI204_DSV_Team12**.

Repository: https://github.com/nivetha486/24ADI204_DSV_Team12

The repository currently contains project-related presentation/data files and a README describing the project, dataset domain, objectives, and Python libraries used. This establishes the version-control foundation for the weekly workflow and later submission of notebooks, scripts, reports and supporting files.

Week 1 GitHub activities

- Created/established the team repository for the project.
- Identified the repository as the central location for project documentation and source files.
- Prepared the project structure for future weekly additions such as notebooks, datasets, reports and code.
- Linked the project direction and dataset description with the repository README.

7. Introduction to Pandas and NumPy

Week 1 also introduces the Python libraries that form the base of the data-processing workflow. **Pandas** will be used for tabular data loading, inspection, cleaning and manipulation, while **NumPy** will support numerical operations and array-based computation.

Planned/introduced operations for the project

- **Pandas:** DataFrame creation, loading CSV data, viewing records, checking dimensions, inspecting columns and data types.
- **Pandas:** Basic selection, filtering and preparation of columns for later analysis.
- **NumPy:** Numerical arrays and basic numerical operations required during data analysis.
- These tools will support the Week 2 'Know Your Data' stage, where the dataset structure and types are examined in detail.

8. Week 1 Alignment with the Rule Book

The rule book defines Week 1 as **Team Formation & Data Hunting**, including selection of the dataset, setting up GitHub, and an introduction to Pandas/NumPy. The team's activities during this week directly correspond to those milestones.

Rule Book Week 1 Requirement	Team Work Completed
Team Formation	Four-member Team 12 formed and project topic finalized.
Data Hunting / Dataset Selection	Comprehensive Autoimmune Disorder Dataset selected for healthcare-focused analysis.
GitHub Setup	Team GitHub repository established for version control and documentation.
Pandas / NumPy Introduction	Initial familiarization with Pandas for tabular data and NumPy for numerical computation.

9. Project Roadmap from Week 1

The Week 1 decisions establish the starting point for the remaining Hack-a-BIT timeline. The team will progressively move from understanding the raw dataset to cleaning, exploratory analysis, feature engineering, dimensionality reduction and finally an interactive narrative dashboard.

Week	Planned Milestone	Project Application
1	Team Formation & Data Hunting	Team, topic, dataset and GitHub foundation
2	Know Your Data	Load dataset; inspect dimensions, types and structure
3	Cleaning Sprint	Missing values and outlier detection
4	EDA Deep Dive	Univariate, bivariate and distribution analysis
5	Mid-Review / Data Audit	Present cleaned dataset and initial findings
6	Feature Engineering	Scaling, normalization and categorical encoding
7	Dimensionality Reduction	Feature selection and PCA
8	Dashboard Construction	Build narrative visualization/dashboard
9	Final Hack-a-BIT Expo	Final presentation, viva and submission

10. Expected Outcomes

- A structured understanding of the autoimmune disorder dataset and its clinical/demographic attributes.
- Clear identification of the variables that are most useful for disease, symptom, biomarker and demographic analysis.
- A clean and well-documented data-processing workflow that can be extended in later weeks.
- Visual evidence of disease patterns and relationships suitable for an interactive dashboard.
- A final visual storytelling product that connects context, findings and actionable insights.

11. Week 1 Deliverables

- Project topic and problem direction finalized.
- Four-member team formed.
- Comprehensive Autoimmune Disorder Dataset selected.
- Initial dataset profile documented from the project materials.
- GitHub repository established for version control and documentation.
- Initial Pandas and NumPy workflow identified for subsequent data exploration.
- Week 2 work prepared: dataset loading, structural inspection and first-impression analysis.

12. Conclusion

Week 1 established the foundation of Team 12's Data Science and Visualization project on autoimmune disorders. The team was formed, the healthcare dataset was selected, the project scope was defined, and a GitHub repository was established for collaborative version control. The dataset provides demographic, symptom, laboratory and biomarker information that can support meaningful visual analysis. The team also began familiarization with Pandas and NumPy, preparing for the next stage of the Hack-a-BIT timeline.

The project will now proceed to Week 2, where the selected dataset will be loaded and systematically inspected to understand its dimensions, column structure, data types and initial data quality.

13. References / Project Materials

- DATA-VIZ HACK-A-BIT (2026–2027) Design Competition Rule Book — Department of Artificial Intelligence and Data Science.
- Autoimmune Disorder Data Analysis and Visualization — Team project presentation.
- Team GitHub repository: https://github.com/nivetha486/24ADI204_DSV_Team12
- Dataset source/reference: Comprehensive Autoimmune Disorder Dataset (Kaggle), as identified in the project presentation and repository README.

Project Status: WEEK 1 COMPLETED